

Mena Karam

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Education

11/2020 – Present Dortmund, Germany	Master of Science in Automation and Robotics <i>Technische Universität Dortmund</i> Current Grade: 1.4 Thesis completed, grade: 1.0 Degree expected: October 2026 Relevant Courses: 3D Computer Vision, Mobile Robots, Modeling and Control of Robotic Manipulators, Machine Learning in Robotics, Remote Sensing, Control Theory and Applications.
09/2015 – 07/2020 Alexandria, Egypt	Bachelor of Science in Mechanical Engineering <i>Arab Academy for Science, Technology and Maritime Transport (AASTMT)</i> Final Grade: 1.0 (German Scale: max: 1.0, min: 4.0) Relevant Courses: Robotics, Mechatronics Systems, Modern Control Engineering, Microcontroller, Automation of Mechanical Systems, Signal Processing, Instrumentation and Measurements.

Selected Research and Robotics Projects

2025	Master's Thesis: Coupled Deblurring and 3D Reconstruction from Image Sequences <i>TU Dortmund, Germany</i> Supervisor: Prof. Dr. rer. nat. habil. Christian Wöhler Grade: 1.0 - Developed a spatially variant deblurring algorithm based on depth-from-defocus and optical geometry for microscopic lunar-regolith image sequences, using MATLAB and Python. - Estimated local blur from circle-of-confusion measurements derived from 3D reconstructions produced by Python-supported Agisoft Metashape SfM-MVS. - Evaluated Wiener and Richardson-Lucy deconvolution with spatially varying PSFs, quantifying effects on image restoration and 3D reconstruction quality. - Achieved 3D reconstructions with increased areal point density and improved spatial resolution, and developed an end-to-end operational workflow for potential use by CAM-M science team.
2020	Bachelor's Thesis: RoboCup Major Teen-Size Humanoid Robot <i>AASTMT College of Engineering, Egypt</i> Awarded Best Thesis with 1.0 grade Project lead. - Derived forward and inverse kinematics and implemented joint-trajectory planning using MATLAB Robotics Toolbox. - Developed ZMP-based dynamically stable gait generation and implemented state-space and model-predictive controllers in MATLAB/Simulink for bipedal locomotion. - Validated multi-DOF humanoid motion in Simscape Multibody and implemented real-time Python/OpenCV perception for HSV segmentation, contour detection, and shape recognition. - Designed MATLAB-ROS interfaces for planned on-robot deployment, with validation remaining at simulation due to pandemic laboratory restrictions. - Assisted in Finite element analysis in SolidWorks under dynamic loading conditions.
2022	3D Models of Planetary Surfaces in Virtual Reality ☑ <i>TU Dortmund, Germany</i> - Developed 3D terrain reconstruction workflow from multi-camera NASA Perseverance imagery based on calibrated and uncalibrated stereo, SfM-MVS, and bundle adjustment. - Implemented workflow using MATLAB, Python, OpenCV, and Agisoft Metashape Pro, and processed multi-view point clouds using ICP in CloudCompare and MeshLab. - Integrated reconstructed terrain models into interactive Unity VR environments for spatial inspection and visualization, and coordinated work in a self-directed, multidisciplinary team.
2019	RoboCup Omniwheel soccer robot <i>AASTMT College of Engineering, Egypt</i> - Applied holonomic kinematic modeling and rigid-body dynamics for autonomous omnidirectional navigation, and integrated ultrasonic, IR, LDR, magnetometer, encoder, and camera sensing for real-time perception and ball tracking. - Implemented a sensor-triggered finite-state machine encoder-feedback closed-loop PWM motor control with UART/SPI communication on a modular embedded architecture.

- 2018 **Quadruped Robot**
AASTMT College of Engineering, Egypt
- Designed quadruped leg kinematics in MATLAB for creep-gait, statically stable locomotion, and integrated rigid-body statics, stress analysis in SolidWorks, and servo control for a prototype.
- 2021 **ROS-Based Manipulator Motion Control — Course Project**
TU Dortmund University
- Implemented inverse kinematics and planned joint-space motions for a simulated UR10 manipulator using MATLAB Robotics System Toolbox, ROS, Gazebo, and RViz.
 - Generated point-to-point and multi-waypoint trajectories; monitored joint position and velocity feedback through ROS topics.
 - Applied ROS publisher–subscriber communication and action-client control for non-blocking trajectory execution with feedback, status and tolerance monitoring.

Practical Experience

- 07/2019 – 08/2019 **Training in Automotive and Electromobility Field**
 Gaggenau, Germany *Carl Benz Schule*
- Completed hands-on training in automotive and electromobility systems including high-voltage safety and lithium-ion battery-management principles.
 - Applied sensor fusion, real-time signal processing, and embedded control concepts in electric/hybrid vehicle contexts.
- 09/2018 **Training in Intelligent Machines and Robotics Application**
 Preston, United Kingdom *University of Central Lancashire*
- Worked with sensors, actuators, microcontrollers, NI myRIO, DAQ and signal conditioning for closed-loop control in LabVIEW.
 - Developed computer vision pipelines using histogram analysis, ROI extraction and HSV/RGB filtering, and integrated digital/analog I/O and PWM in autonomous search-and-rescue system.
 - Awarded Best Group Project for the “Search and Rescue Autonomous Robot” project.

Technical Skills

- Perception and 3D Vision:** Optical image formation, depth-from-defocus, image processing, stereo vision, SfM/MVS, bundle adjustment, point-cloud registration and refinement; OpenCV, Agisoft Metashape Pro, CloudCompare, MeshLab.
- Mobile Robotics, Navigation and ROS:** ROS, Gazebo, RViz, publisher–subscriber and action-client communication; odometry, LiDAR-based localisation, mapping, and scan-matching foundations, Monte Carlo localisation, occupancy-grid mapping; PRM/RRT-based planning, reactive obstacle avoidance, path following, and online trajectory optimisation.
- Humanoid Robotics and Control:** Forward/inverse kinematics, joint-trajectory planning, ZMP-based gait generation, state-space and model-predictive control, finite-state machines, MATLAB–ROS integration, and multibody simulation.
- Programming and Engineering Tools:** Python (intermediate–advanced), MATLAB/Simulink (advanced), C++, Git, Linux, LabVIEW, Unity, SolidWorks, Solid Edge, AutoCAD, Autodesk Eagle, and FEA.
- Embedded Systems:** Multi-sensor integration, encoder-feedback PWM motor control, microcontrollers, UART/SPI, PCB design, signal conditioning, DAQ, and NI myRIO.
- Machine Learning:** PyTorch and TensorFlow; foundational practical experience with data preprocessing, neural-network models, reinforcement-learning concepts, and learning-based perception.

Languages

English — C2 | **Arabic** — Native | **German** — B1 (currently improving)

Interests

Playing musical instruments — Synthkeyboard, bass and guitar.

Reading — Classical literature, history, arts, and Astrophysics.

Film Photography